# Ashwin Chakravartula

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## Education

**RWTH Aachen**, M.Sc. Robotic Systems Engineering — Oct 2023 – Present

**Birla Institute of Technology and Science, Pilani** , B.Sc Mechanical Engineering — Aug 2019 - Jun 2023

## Experience

**Working Student**, Siemens AG – Munich — Aug 2026 – Present

- Implemented a robust perception pipeline using FoundationPose for precise 6 DOF target pose estimation, validating physical camera offsets via AprilTag calibration.
- Designed CAD models for industrial trays and 3D-printed custom camera wrist mounts, utilizing AprilTags to validate physical offsets and ensure precise calibration.
- Bridged the Sim2Real gap for Unitree G1 humanoids by transferring simulated Curobo trajectories to physical hardware, implementing feed-forward gravity compensation to reduce arm-sagging positional error by over 60%.

**Working Student**, German Aerospace Center – Munich — Jan 2026 – Present

- Developed model-based synthetic surface dataset generation pipelines for offline trajectory planning of the SARA robotic platform.
- Implemented prompt-conditioned diffusion models such as Point-E and Shap-E for 3D point cloud surface generation using PyTorch and Open3D.
- Designed evaluation metrics for mesh fidelity and traversability to improve downstream motion-planning robustness.

**Student Research Assistant**, Institute for Advanced Mining Technologies – Aachen — July 2025 – Jan 2026

- Integrated stereo vision perception pipelines on NVIDIA Jetson Orin using NVIDIA Isaac ROS for real-time depth estimation.
- Developed and containerized ROS2 perception nodes using Docker for edge deployment and reproducible testing.
- Implemented point cloud filtering and segmentation pipelines for environment perception.

**Research Project**, Institute of Industrial Engineering and Ergonomics – Aachen — Apr 2025 – Dec 2025

- Developed a VR-based teleoperation framework for robotic grinding applications using Unity and ROS2.
- Built a real-time digital twin pipeline integrating haptic joint control with motion-planning algorithms.
- Benchmarked IK based trajectory generation against direct joint mapping for motion smoothness and operator control stability.
- Engineered workspace parameterization and collision-aware motion constraints for scalable deployment.

**Student Research Assistant**, Mechatronics and Mobile Propulsion Institute – Aachen — Jan 2024 – Sept 2024

- Evaluated and benchmarked global and local path-planning strategies for navigation efficiency and obstacle avoidance.
- Integrated occupancy-grid SLAM and sensor fusion pipelines for localization and navigation stack.

## Skills

**Programming:** Python, C++, C#, MATLAB

**Robotics & Simulation:** ROS/ ROS2, MoveIt2, Gazebo, MuJoCo, Unity 3D, NVIDIA IsaacLab

**Perception & AI:** PyTorch, Open3D, Sensor Fusion, Point Cloud Processing, BlenderProc

**Tools & Platforms:** Docker, Linux, Git, Jetson Orion, Kubernetes, ArgoCD

**Language Proficiency:** English (Fluent), German (B1)